

MobileViT-Based Traffic Analysis for Digital Twin Optimization

Course: EPL 445 – Digital Video Processing

Interim Presentation #1 (3–5 minutes)

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Date: 06 Apr 2026

1) Scientific goal & question

Goal

Provide intelligent analysis of urban traffic video to support data-driven decision making in a “digital twin” context.

Interesting question

How can we estimate traffic density and vehicle categories in near real-time using a lightweight transformer backbone?

Output (for the digital twin)

- Per-frame detections and vehicle class (car/bus/truck/motorbike, etc.)
- Traffic density per lane/region of interest (ROI)
- Aggregated counts over time (minute-by-minute time series)

2) Methodology (MobileViT-based pipeline)

Processing pipeline (high level)

- Ingest video stream (file or RTSP) → sample frames
- MobileViT backbone for compact feature extraction
- Detection head (lightweight) → vehicle boxes + class
- Optional tracking (SORT/ByteTrack-lite) → stable counts
- ROI aggregation → density + per-class counts
- Expose results via a web dashboard / API

Why MobileViT?

- Lightweight transformer designed for mobile-friendly vision
- Good accuracy/efficiency trade-off vs. heavier ViTs
- Fits the “mobile video processing & analysis” spirit

Implementation
Python + OpenCV
PyTorch/torchvision
Web: FastAPI + simple UI

3) Resources & feasibility

Software

- Python, OpenCV (frame I/O + basic preprocessing)
- PyTorch + torchvision (model + inference)
- Optional: PyTorchVideo (video utilities)
- scikit-learn (metrics computation)

Models

- Start from pre-trained weights (ImageNet backbone)
- Fine-tune detection head on a traffic dataset
- Fallback: use fully pre-trained detector for first runs

Hardware & expected timing

- GPU recommended for near real-time inference
- Target: ≥ 10 FPS on resized frames (e.g., 640×360) on a mid-range GPU
- CPU-only mode supported for demo (lower FPS)
- Web deployment: one GPU worker + REST API

Feasibility strategy

Deliver first working pipeline quickly using pre-trained weights, then improve accuracy with fine-tuning + tracking.

4) Evaluation plan & milestones

Datasets (candidate options)

- Traffic-focused: UA-DETRAC, BDD100K (videos), AI City Challenge
- Tracking-oriented: VOT (for tracking module evaluation)
- If needed for quick prototyping: any short YouTube traffic clips

Metrics

- Classification-style: Accuracy, Sensitivity, Specificity, F-Score
- ROC curve + AUC (where applicable)
- Detection: Precision/Recall and mAP
- Counting: MAE / MAPE vs. ground truth counts

Milestones

06 Apr: Interim #1 (plan + resources)
By 23 Apr: “first runs” + metrics on test/eval set
Early May: improve tracking + ROI counting
25 May: final demo (web app) + report submission